

Data Analytics Report

Source file: train.csv

1. Dataset Overview

Rows (after cleaning)	9,800
Columns	18
Missing values handled	11
Duplicates removed	0
Type conversions	0

Preview of cleaned data

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID
1	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520
2	CA-2017-152156	08/11/2017	11/11/2017	Second Class	CG-12520
3	CA-2017-138688	12/06/2017	16/06/2017	Second Class	DV-13045
4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335
5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335
6	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710
7	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710
8	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710
9	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710
10	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710
11	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710
12	CA-2015-115812	09/06/2015	14/06/2015	Standard Class	BH-11710

2. Data Cleaning Summary

- Filled 11 missing value(s) in 'Postal Code' with median (58103.00).

3. Exploratory Data Analysis

3.1 Descriptive Statistics

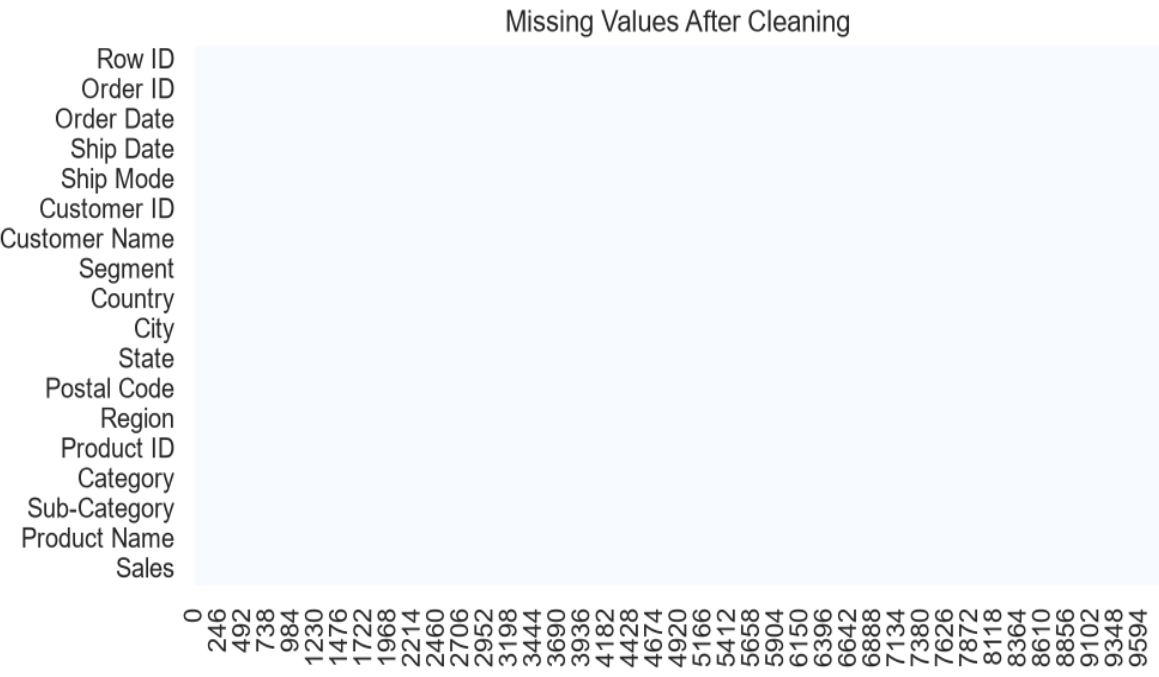
count	unique	top	freq	mean	std
9,800.00				4,900.50	2,829.16
9800	4922	CA-2018-100111	14		
9800	1230	05/09/2017	38		
9800	1326	26/09/2018	34		
9800	4	Standard Class	5859		
9800	793	WB-21850	35		
9800	793	William Brown	35		
9800	3	Consumer	5101		
9800	1	United States	9800		
9800	529	New York City	891		
9800	49	California	1946		
9,800.00				55,276.50	32,023.37
9800	4	West	3140		
9800	1861	OFF-PA-10001970	19		
9800	3	Office Supplies	5909		

3.2 Correlation Matrix

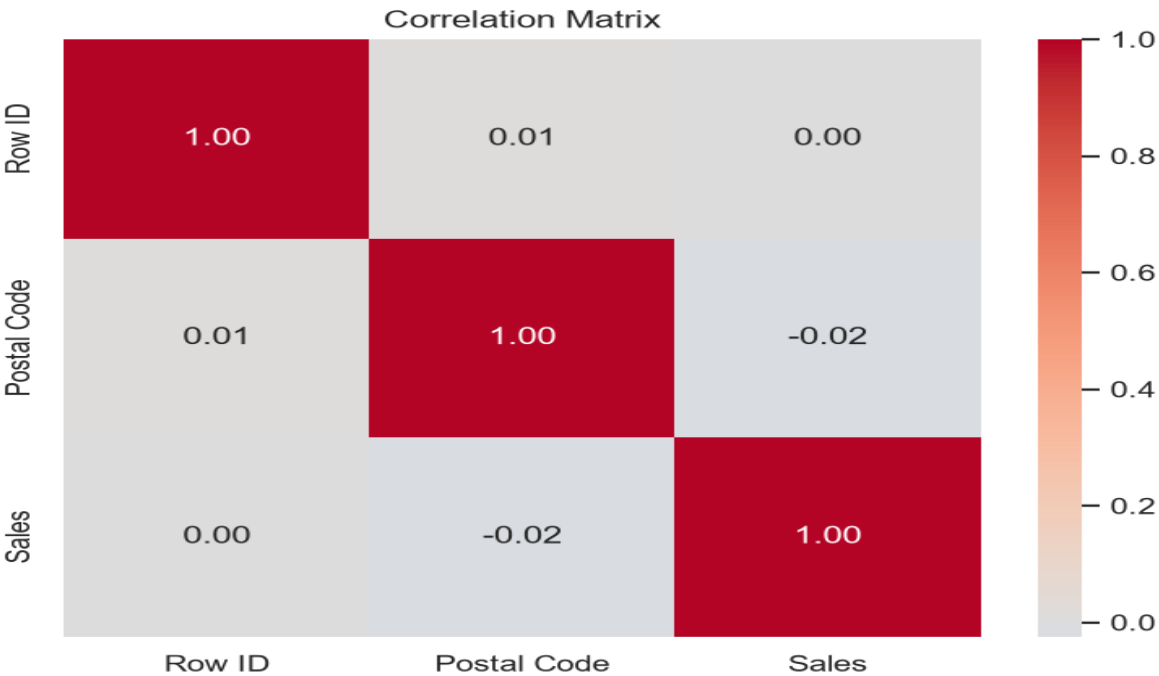
Row ID	Postal Code	Sales
1.00	0.01	0.001151
0.01	1.00	-0.02
0.001151	-0.02	1.00

3.3 Visualizations

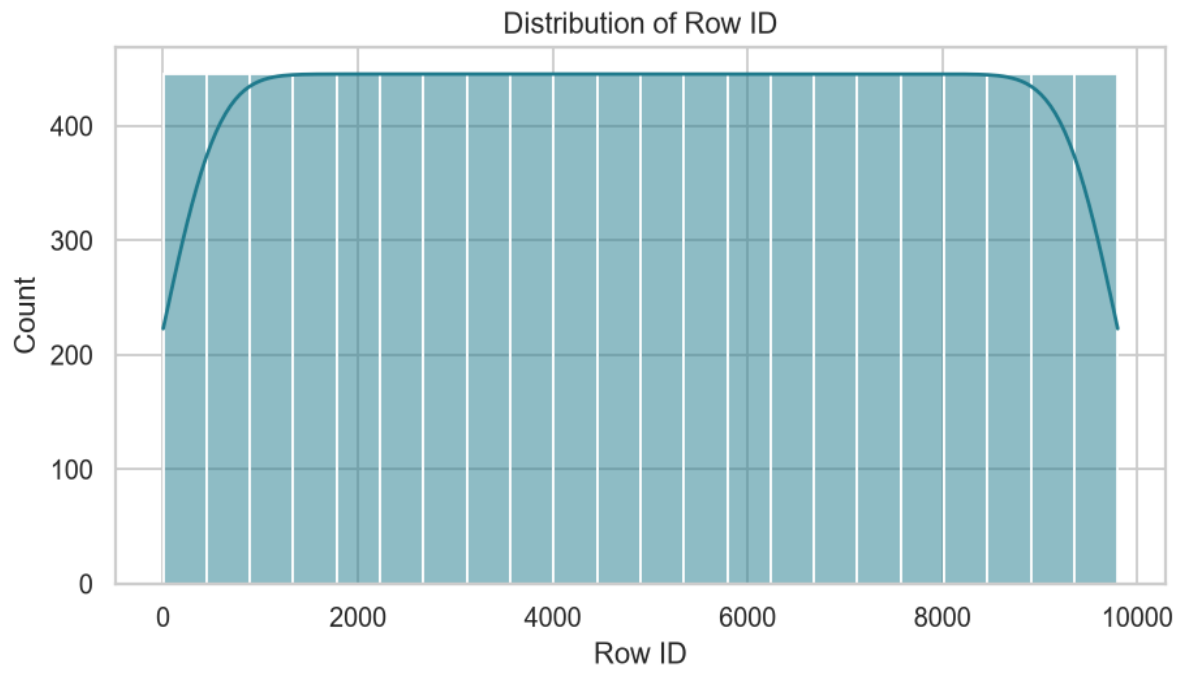
Missing Values



Correlation Heatmap



Distribution Row Id



Count Order Id



4. Business Insights & Recommendations

- The dataset contains 9,800 rows across 18 columns.
- Numeric columns analyzed: Row ID, Postal Code, Sales.
- No strong correlations ($|r| \geq 0.6$) were found among numeric columns.
- 'Row ID' averages 4,900.50 with high variability (std 2,829.16).
- 'Postal Code' averages 55,276.50 with high variability (std 32,023.37).
- 'Sales' averages 230.77 with high variability (std 626.65).
- The most common 'Order ID' is 'CA-2018-100111' (14 records).
- Recommendation: prioritize the strongest correlated variables for predictive modeling.
- Recommendation: investigate high-variability columns for data quality or segmentation opportunities.
- (LLM insight generation skipped: Error code: 401 - {'error': {'message': 'Incorrect API key provided: AQ.Ab8RN*****Hdug. You can find your API key at https://platform.openai.com/account/api-keys.', 'type': 'invalid_request_error', 'param': None, 'code': 'invalid_api_key'}})